

Azure Administrator (AZ-104) Course Syllabus

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Course Overview

This syllabus lists every module of the course in teaching order, with the topics covered in each. It is generated from the course page, so the two cannot disagree.

Module 1: Exam Strategy, Subscription Setup & Foundations Review

Schedule: Week 1

How the AZ-104 exam scores you, the question patterns Microsoft uses (case studies, drag-and-drop, multiple-choice, multiple-response), test-day logistics. Then a foundations review — Azure subscription / management group hierarchy, Microsoft Entra ID basics, RBAC scope and inheritance. By the end of week 1 every student has a personal Azure sandbox subscription and has done a 50-question diagnostic exam.

Topics

- AZ-104 exam structure and domain weighting
- Microsoft question patterns — case studies, drag-and-drop
- Test-day logistics — Pearson VUE
- Azure subscription hierarchy
- Microsoft Entra ID (renamed Azure AD)
- RBAC scope and inheritance
- First diagnostic exam

Module 2: Azure Fundamentals Refresher — Portal, CLI, PowerShell & Bicep

Schedule: Week 1

AZ-104 is an administrator exam, and it assumes fluency in the tools before it tests judgement about them. This week gets you comfortable in all four surfaces you will be examined on: the portal, Azure CLI, Azure PowerShell, and ARM templates with Bicep.

That breadth matters practically — the exam and the job both mix them, and a candidate who only knows the portal stalls the moment a question shows a command. You cover the resource hierarchy, resource groups and what they actually scope, tags, locks and the deployment model underneath everything, then deploy the same resource four ways so the equivalences are concrete.

Topics

- Azure regions, availability zones and paired regions
- Management groups, subscriptions and resource groups
- The Azure Resource Manager deployment model
- The portal, Cloud Shell and where each is faster
- Azure CLI syntax and common commands
- Azure PowerShell and the Az module
- ARM templates and Bicep
- Tags, resource locks and naming standards
- Deploying the same resource four ways
- Reading Microsoft Learn documentation efficiently

Module 3: Microsoft Entra ID in Depth — Users, Groups, RBAC & Conditional Access

Schedule: Week 2

The identity foundation the largest scored domain rests on. Users, groups and administrative units; dynamic group membership with the rule syntax the exam shows you; guest accounts and external collaboration; and self-service password reset.

RBAC gets the most time because scope is where candidates lose marks: assignments inherit down the hierarchy, deny assignments behave differently from Azure Policy, and a role assigned at the wrong scope is the single most common wrong answer in this domain. Conditional Access, MFA and Identity Protection close the module, along with the built-in roles you are expected to recognise by what they permit rather than by name alone.

Topics

- Users, groups and administrative units
- Dynamic group membership rules
- Guest users and external collaboration
- Self-service password reset and writeback
- RBAC roles, scope and inheritance
- Built-in roles and what each actually permits
- Custom role definitions
- Deny assignments versus Azure Policy
- Conditional Access policies and MFA
- Entra ID Protection and risk-based policies

Module 4: Domain 1 — Manage Azure Identities and Governance (~20–25%)

Schedule: Weeks 2–3

Identity is the largest exam domain. Cover Microsoft Entra ID in depth — users, groups, dynamic groups, external identities (B2B / B2C basics), MFA, Conditional Access, Privileged Identity Management. RBAC — built-in and custom roles, scope, ABAC. Subscriptions — moving resources, transferring billing. Azure Policy — definitions, initiatives, exemptions, compliance. Plus Cost Management — budgets, alerts, recommendations.

Topics

- Microsoft Entra ID users, groups, dynamic groups
- External identities (B2B / B2C)
- MFA and Conditional Access
- Privileged Identity Management
- RBAC — built-in, custom, scope
- Subscriptions and resource moves
- Azure Policy and Initiatives
- Cost Management — budgets, alerts

Module 5: Governance — Policy, Cost Management & Resource Organisation

Schedule: Week 3

The other half of Domain 1, and the part that separates an administrator from someone who can click through the portal. Azure Policy — definitions, initiatives, effects such as deny, audit and deployIfNotExists, and remediation tasks for resources already out of compliance.

Cost Management covers budgets, alerts, cost analysis and the exports finance teams actually ask for; the pricing and TCO calculators; and reservations and Azure Hybrid Benefit. Resource organisation closes it — management group hierarchy, subscription strategy, tagging enforced by policy rather than by hope, and the move and lock operations the exam tests with specific constraints on what can move where.

Topics

- Azure Policy definitions and initiatives
- Policy effects — deny, audit, append, deployIfNotExists

- Remediation tasks for existing resources
- Policy versus RBAC — different questions
- Budgets, alerts and cost analysis
- Reservations and Azure Hybrid Benefit
- Pricing and TCO calculators
- Management group hierarchy and subscription strategy
- Tag inheritance and enforcement through policy
- Moving resources between groups and subscriptions

Module 6: Domain 2 — Implement and Manage Storage (~15–20%)

Schedule: Week 4

Storage domain. Cover Storage Accounts — kinds (general-purpose v2, BlobStorage, FileStorage), access tiers, redundancy options (LRS / ZRS / GRS / GZRS), authorisation methods (Shared Key, SAS, Entra ID), plus the modern container-by-container access controls. Blob storage — tiers (Hot / Cool / Cold / Archive), lifecycle management, soft delete, immutable storage. File shares — SMB / NFS, AD authentication. Plus Azure File Sync, Storage Explorer, AzCopy.

Topics

- Storage Account kinds and selection
- Access tiers — Hot / Cool / Cold / Archive
- Redundancy — LRS / ZRS / GRS / GZRS
- SAS, Entra ID, Shared Key authorisation
- Blob lifecycle and soft delete
- Azure Files — SMB / NFS
- Azure File Sync, AzCopy

Module 7: Storage Security, Replication & Azure Files

Schedule: Week 4

The depth behind Domain 2, where the exam's questions are almost always about choosing a redundancy option or an access mechanism. The replication tiers — LRS, ZRS, GRS, GZRS and their read-access variants — with the durability and failover behaviour of each, because the correct answer is decided by a stated requirement rather than by preference.

Access control covers shared keys, shared access signatures with stored access policies, and Entra-based authorisation, which is the recommended and frequently correct choice. Azure Files with Active Directory authentication, Azure File Sync, blob lifecycle management and immutable storage close the module, along with the tools — AzCopy, Storage Explorer, Import/Export — the exam expects you to distinguish.

Topics

- Storage account types and performance tiers
- LRS, ZRS, GRS, GZRS and read-access variants
- Blob access tiers and lifecycle management
- Immutable storage and legal holds
- Shared keys, SAS tokens and stored access policies
- Entra ID authorisation for blob and queue
- Azure Files and AD authentication
- Azure File Sync and cloud tiering
- AzCopy, Storage Explorer and Import/Export
- Firewalls, private endpoints and network rules

Module 8: VM Availability, Scale Sets & Backup

Schedule: Week 5

The resilience half of the compute domain. Availability sets with fault and update domains, availability zones, and the SLA each arrangement actually carries — the exam asks for the configuration that meets a stated SLA, and the numbers matter.

Virtual Machine Scale Sets cover autoscaling rules, scaling profiles and upgrade policies. Azure Backup covers Recovery Services vaults, policies, retention and restore options; Azure Site Recovery covers replication and failover for disaster recovery. Disk management, encryption and resizing close the module, alongside the maintenance and reboot behaviours that scenario questions turn on.

Topics

- Availability sets, fault and update domains
- Availability zones and zone-redundant deployment
- SLA by configuration — the numbers the exam uses
- Virtual Machine Scale Sets and autoscale rules
- Scaling profiles and upgrade policies
- Managed disk types, resizing and encryption
- Recovery Services vault and backup policies
- Restore options and retention
- Azure Site Recovery and failover
- Maintenance, reboots and planned events

Module 9: Domain 3 — Deploy and Manage Compute Resources (~20–25%)

Schedule: Weeks 5–6

Compute domain. VMs — sizing, availability sets vs zones vs scale sets, custom images via Azure Image Builder, Azure Bastion, plus VM extensions and Update Manager. App Service — plans, slots, deployment, scaling. Container services — Azure Container Instances, Azure Container Apps, AKS basics (Automatic mode covered as the 2026 default). Plus Backup and Site Recovery for compute.

Topics

- VM sizing and pricing models
- Availability sets, zones, scale sets
- Azure Image Builder and shared image gallery
- Azure Bastion for secure RDP / SSH
- App Service — plans, slots, scaling
- Azure Container Instances and Container Apps
- AKS basics (Automatic mode)
- Azure Backup and Site Recovery

Module 10: App Service, Containers & Serverless Compute

Schedule: Week 6

The rest of Domain 3, covering the compute options that are not virtual machines. App Service plans and tiers, deployment slots and slot swapping — which appears constantly in scenario questions about zero-downtime release — scaling, custom domains and TLS binding.

Containers cover Azure Container Instances for short-lived workloads, Container Apps, and the level of AKS the exam actually expects, which is provisioning and scaling rather than deep Kubernetes operations. Azure Functions and consumption plans close the module, along with the selection question the exam keeps asking: given this workload and this constraint, which compute service is correct.

Topics

- App Service plans, tiers and scaling
- Deployment slots and slot swapping
- Custom domains, TLS binding and certificates

- App Service networking and access restrictions
- Azure Container Instances and Container Apps
- Azure Container Registry
- AKS at the level AZ-104 tests
- Azure Functions and hosting plans
- Choosing a compute service from a constraint
- Migrating a VM workload to a managed service

Module 11: Domain 4 — Implement and Manage Virtual Networking (~20–25%)

Schedule: Weeks 6–7

Networking domain. VNet design — subnets, NSGs, ASGs, route tables, service endpoints vs Private Endpoints (the modern default). Cross-VNet — peering, VPN Gateway, ExpressRoute, Virtual WAN. Load balancing — Azure Load Balancer (basic + standard), Application Gateway with WAF, Azure Front Door, Traffic Manager — and the discipline of choosing the right one for the scenario. Plus Azure DNS / Private DNS Zones.

Topics

- VNet, subnets, NSGs, ASGs
- Route tables and UDR
- Service Endpoints vs Private Endpoints
- VNet Peering
- VPN Gateway and ExpressRoute
- Azure Virtual WAN
- Load Balancer (basic / standard)
- Application Gateway with WAF
- Front Door vs Traffic Manager
- Azure DNS / Private DNS Zones

Module 12: Network Security, Load Balancing & Private Connectivity

Schedule: Week 7

The depth behind the networking domain, and the area candidates most often underestimate. Network security groups and application security groups, effective security rules, and reading why traffic is actually being blocked — which the portal will tell you if you know where to look.

Load balancing covers the four services the exam expects you to distinguish: Azure Load Balancer, Application Gateway with WAF, Traffic Manager and Front Door, each with the layer it works at and the scenario it fits. Private connectivity covers service endpoints against private endpoints, Private Link, VNet peering and its transitivity limits, VPN Gateway and ExpressRoute, plus Azure DNS and private zones.

Topics

- NSGs, ASGs and effective security rules
- Diagnosing blocked traffic
- Azure Load Balancer — basic and standard
- Application Gateway and WAF
- Traffic Manager and Front Door
- Choosing between the four load-balancing services
- Service endpoints versus private endpoints
- Private Link and private DNS integration
- VNet peering and transitivity limits
- VPN Gateway, ExpressRoute and hybrid connectivity

Module 13: Domain 5 — Monitor and Maintain Azure Resources (~10–15%)

Schedule: Week 8

Observability and maintenance. Azure Monitor — metrics, logs, KQL queries, Log Analytics workspaces, action groups, alert rules. Application Insights for application-level telemetry. Backup — Recovery Services Vault, backup policies, cross-region restore. Plus Azure Resource Health, Service Health, Service Manager.

Topics

- Azure Monitor — metrics, logs, alerts
- Log Analytics and KQL
- Application Insights
- Recovery Services Vault and Backup policies
- Site Recovery for DR
- Resource Health and Service Health
- Azure Resource Manager templates

Module 14: Mock Exam 1, Scenario Walkthroughs & Gap Closure**Schedule:** Week 9

First full-length timed mock exam (60 questions, 100 minutes) under real exam conditions, including the Pearson VUE-style case-study format. Detailed answer review focusing on why wrong options are wrong. Plus scenario walkthroughs of common AZ-104 architectures — multi-region web app, hybrid identity, BFSI compliance setup.

Topics

- Mock exam 1 — full timed
- Detailed answer review
- Scenario walkthroughs
- Gap-closure plan based on mock score

Module 15: Mock Exam 2, Booking & Test-Day Prep**Schedule:** Week 10

Second full-length mock exam under real exam conditions — students who score 75–85% typically pass the real exam first attempt. Final gap-closure on weak domains, exam-booking guidance, test-day mental-prep, plus retake strategy if first attempt fails.

Topics

- Mock exam 2 — full timed
- Final gap-closure
- Pearson VUE booking
- Test-day strategy
- Retake strategy

Module 16: AI-Assisted Development Workflow**Schedule:** Final Week

The skill every 2026 hiring panel now probes for — building real work with AI in the loop, responsibly. Learn to drive AI assistants (GitHub Copilot, Claude, Cursor, and IDE-native AI) to scaffold and accelerate the tools and stack this course covers, generate tests, explain and refactor unfamiliar code, and cut the boilerplate — while keeping you firmly in control of every decision. Heavy focus on guardrails: reviewing each AI suggestion, spotting hallucinated APIs or wrong answers, and handling licensing and data-privacy concerns. Close with a mini-project that takes a deliverable end-to-end using an AI-assisted workflow, then fold the same tooling into version control and everyday team practice.

Topics

- AI assistants — GitHub Copilot, Claude, Cursor, IDE-native AI

- Effective prompting for this course's stack — scaffolding, boilerplate, config
- AI-assisted test generation and coverage
- Explaining, refactoring, and modernising unfamiliar code with AI
- AI-driven review, error detection, and quality checks
- Generating and maintaining documentation with AI
- AI debugging — interpreting errors, logs, and failing output
- Guardrails — reviewing output, avoiding hallucinations, licensing & data privacy
- Team workflow — AI in the editor, in reviews, and in delivery pipelines
- Mini-project — a deliverable built end-to-end with an AI-assisted workflow

Enquire or book a free demo session. Archer Infotech, Flat No. 12, Divyadarshan Housing Society, behind Kothrud Bus Stand Road, Kothrud, Pune 411038. Phone +91 9850 678451. Email info@archerinfotech.in. Web archerinfotech.in.